

Full-Stack & AI Engineer · Agentic LLM Systems · Production Web & Mobile Apps

Software engineer with 2+ years shipping full-stack products end to end — Next.js, React Native and Spring Boot / FastAPI on AWS — and building the AI layer on top: MCP servers and tool-using Claude pipelines with schema-validated outputs, guardrails and human-in-the-loop review. M.Sc. in Artificial Intelligence; AWS Certified Solutions Architect.

Experience

Hanz

Stuttgart, Germany (remote)

Software Engineer

Sep 2024 – present

- Built a production voice-first reporting system for multilingual construction crews: workers speak their daily report and a FastAPI AI service uses Claude tool use (schema-as-tool contract) to extract structured tasks linked to exact database records, grounded by project context ranked in SQL (context injection over RAG for small, closed entity sets).
- Made the LLM output safe to act on: layered checks (Pydantic schema, validators, verbatim ID verification) catch every invented reference, and an adaptive human-in-the-loop review wizard asks workers only what the model could not resolve — no AI output is saved without confirmation.
- Redesigned the extraction prompt from a 5,466-word rule list into principles and reasoned examples (78% shorter) and moved checkable invariants into code, lowering latency and improving extraction accuracy.
- Designed and implemented an MCP server that bridges the REST backend with LLM clients, exposing 30+ tools with request-scoped authentication so users can query project data in natural language.
- Implemented real-time speech-to-text (Soniox streaming ASR) over a server-side WebSocket proxy for low-latency live captioning, with auto-reconnect and local audio backup uploaded to S3; extended the pipeline to Claude vision for handwritten site sheets.
- Led and shipped the migration of the production backend from Kotlin/Ktor to Java/Spring Boot — architected a package-per-feature rewrite (JPA/Hibernate, Spring Security + JWT, Flyway) across 25+ domains, preserving the existing REST contract so mobile clients required no changes.
- Cut average request-handling time by 50%+ on backend services by moving ORM-layer logic into PostgreSQL stored procedures (PL/pgSQL).
- Built the Next.js / React web app for managing construction projects and generating invoices, and the Expo / React Native mobile app (44 languages), owning the full release pipeline to the App Store and Google Play.
- Containerized services with Docker and GitHub Actions deployments, and migrated Supabase storage to AWS S3 (encrypted, pre-signed URLs), lowering storage costs and tightening access control.

Vanrise

Beirut, Lebanon

Full-stack Developer Intern

Aug – Sep 2024

- Built a full-stack application using .NET MVC Architecture, Angular, ADO.NET, MS SQL, and Bootstrap.

Projects

CliniKit — AI Patient-Intake Agent & No-Show Predictor — Sep 2026

Built a Claude-powered intake assistant (Python, FastAPI) that turns messy patient messages into Pydantic-validated intents and slots via forced tool use, with a deterministic guardrail that only books, reschedules, or cancels when

required details are present and the patient explicitly confirmed. Trained a scikit-learn no-show prediction model on 3,000 real appointment records, evaluated on ROC-AUC and no-show recall for the imbalanced target, and designed its AWS deployment (Bedrock, Lambda, SageMaker).

ArcVault Intake-Triage — Apr 2026

Built an agentic n8n workflow that ingests unstructured customer messages, classifies them by category and priority, extracts entities, routes them to the correct team queue, and flags low-confidence or high-risk cases for human escalation.

Habit-tracking Graph Editor — 2026

Node-based habit tracker built with Next.js, React, TypeScript and ReactFlow, with a form-driven side panel (React Hook Form + Zod) for editing node data.

Skills

AI & LLM: Claude API (tool use, structured outputs, vision, prompt caching), MCP server design, agentic workflows, LangGraph, n8n, context engineering, RAG, guardrails & human-in-the-loop, Gemini, OpenAI
ML: scikit-learn, model evaluation (ROC-AUC, recall on imbalanced data)
Languages: Python, TypeScript, JavaScript, Java, Kotlin, SQL
Backend: Spring Boot (JPA/Hibernate, Spring Security, Flyway), FastAPI, Pydantic, Express.js, ASP.NET; REST, WebSockets, JWT, Auth.js
Frontend & Mobile: React, Next.js, React Native, Expo (EAS), Tailwind CSS, shadcn/ui, Material UI, React Hook Form, Zod
Data: PostgreSQL (PL/pgSQL), Supabase, Prisma, Drizzle, MS SQL
Cloud & DevOps: AWS (S3, Lambda, Bedrock, SageMaker), Docker, GitHub Actions, Git, PostHog

Certifications

- Claude Certified Architect — Anthropic — in progress (2026)
- AWS Certified Solutions Architect — AWS — Sep 2026
- Advanced MCP — Anthropic — Apr 2026
- AWS Certified Cloud Practitioner — AWS — Jul 2024
- CS50 — Harvard University — Aug 2024

Education

Lebanese University	Hadat, Lebanon
M.Sc. in Artificial Intelligence (professional track)	2024 – 2026
Bachelor of Science in Computer Science	2021 – 2024
Thesis: <i>Voice-First AI Daily Reporting for Construction Sites — Grounded LLM Extraction with Human-in-the-Loop Review in a Production System (with Hanz, Stuttgart)</i>	

Relevant Coursework: Machine Learning, Deep Learning, NLP, LLMs, Computer Vision, Design Patterns, Parallel Programming, DBMS, Network Security.

Languages

- English** — Bilingual
French — B2 (DELF)
Arabic — Native